



Distributional Semantics

English Literature Semantic Change

DATA SEMANTICS - MSC DATA SCIENCE - 2026

Domain: Project Gutenberg Literary Corpus | 1800–1980

Fayyaz Hussain Shah · 946912

Meeran Meeran · 948445

Università degli Studi di Milano-Bicocca

GIT: <https://github.com/Fayyaz6137/Literature-Semantic-Change>

Responsible AI Usage

Code Debug

Level - Medium -> (Claude - Sonnet 4.6)

Project Structure

Level - Low -> I wanted the project to follow professional standards of file structure and logging hence I took AI's help in learning to do that -> (Claude - Sonnet 4.6).

Visual Enhancements

Level - High -> I was not satisfied by default color scheme hence I took AI's help to create my own color palette for my graphs and use it in my code. (ChatGPT - OpenAI)

Learning Concepts and Theory

The explanation in research papers was not simple enough hence I used AI to explain in lay man's language

What is Distributional Semantics?

"You shall know a word by the company it keeps."

— J.R. Firth, 1957

THE CORE IDEA

Words appearing in similar contexts have similar meanings. Meaning emerges from usage patterns not from definitions.

COSINE SIMILARITY

Measures the angle between two vectors. Score = 1.0 means identical context (nearly synonymous). Score ≈ 0 means unrelated. Our key metric throughout this project.

WORD VECTORS

Train a neural network on text → each word gets a list of ~100 numbers (a "vector"). Similar words point in the same direction in this mathematical space.

WHY IT MATTERS

We never manually define meanings. The corpus defines them. If "awful" starts appearing near "terrible" instead of "sublime", its meaning has changed and we can measure it.

Domain & Research Questions

Project Gutenberg

70,000+ public-domain books

1800 – 1980 | 9 literary

periods

100% free — no licence

required

RQ1

Which words changed meaning most between the 1800s and 1960s?

RQ2

For the top 5 changers what specific shift occurred?

RQ3

How did target words evolve across 9 periods?

RQ4

Are agency adjectives more associated with male names in 19th-century literature?

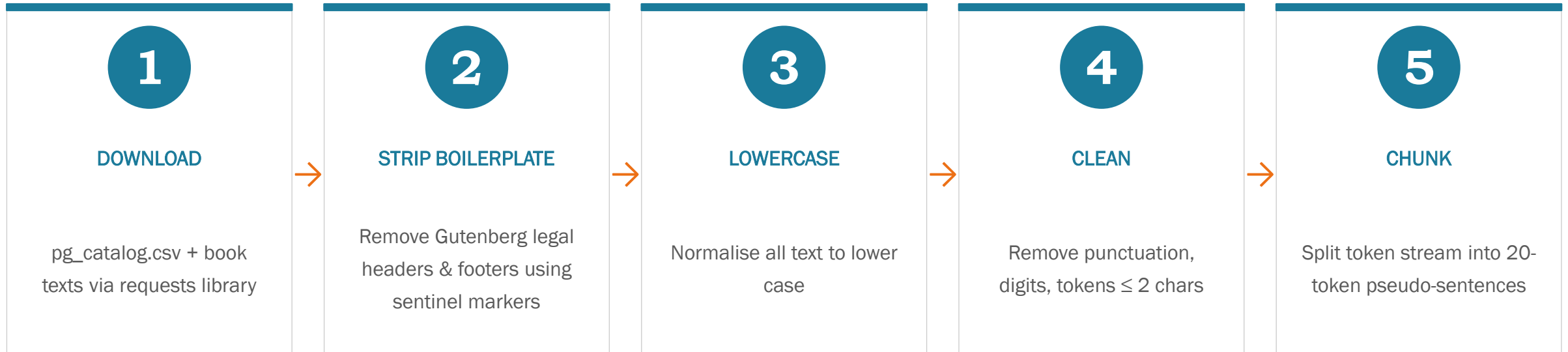
RQ5

Has this gender bias changed from the 1800s to the 1960s?

The Gutenberg Corpus - Data Description

Period	Years	Words as tokens	Size Mb
1800	1800 - 1820	19,219,650	113.1
1820	1820 - 1840	18,448,239	108.5
1840	1840 - 1860	18,600,537	108.5
1860	1860 - 1880	15,808,548	91.3
1880	1880 - 1900	15,184,709	88.4
1900	1900 - 1920	13,943,291	80.8
1920	1920 - 1940	12,671,780	73.4
1940	1940 - 1960	6,800,598	40.9
1960	1960 - 1980	4,343,957	26
TOTAL	1800 - 1980	125,021,309	731

Preprocessing Pipeline



⚠ Stopwords are NOT removed — they provide essential contextual signals for Word2Vec. The sentence "she was a brave and noble woman" trains the association between "brave" and female-gender contexts through the co-occurrence with "she" and "woman". However Stopwords were filtered before analyzing the final results.

Word2Vec - Skip-gram Model

How Skip-gram Works

Context window (size 5)



Centre word: "brave" → predict context: "She", "was", "and", "noble"

After millions of such examples, vectors for "brave" and "noble" end up pointing in similar directions → high cosine similarity → semantically related.

brave [22,66,87,3,4n] n = vector_size

noble [89,45,67,8,22n] n = vector_size

Hyperparameters

vector_size	200	Each word = 200 numbers
window	5	5 words each side
min_count	5	Filter rare words
sg	1	Skip-gram (not CBOW)
epochs	5	Training passes

Why Skip-gram over CBOW? Skip-gram generates more training examples per word and outperforms CBOW on rare vocabulary that is critical for literary corpora with archaic, period-specific terms.

CADE - Compass-Aligned Distributional Embeddings

⚠ The problem: Train Word2Vec separately on 1800s text and 1960s text → two different coordinate systems → cannot compare vectors directly!



✓ The CADE fix: Each slice model is fine-tuned from the compass weights, not initialised randomly. All slices share the same coordinate frame.

```
change(word) = 1 - cosine_similarity( v_1800s(word), v_1960s(word) ) → High score = large  
semantic drift
```


WEAT & SWEAT - Measuring Gender Bias

Word Sets

Agency Traits (T_A)

brave, strong, rational, bold,
independent, ambitious, courageous

Communal Traits (T_C)

gentle, beautiful, emotional, frail,
delicate, modest, tender, meek

Male Names (A₁)

henry, william, edward, charles,
george, thomas, arthur, james

Female Names (A₂)

mary, elizabeth, jane, catherine,
emma, agnes, clara, harriet

How WEAT Works

For each target word t , compute:

$$s(t) = \text{mean_cos}(t, A_1) - \text{mean_cos}(t, A_2)$$

Effect size: $d = \text{mean}(s) / \text{std}(s)$

Positive $d \rightarrow$ targets closer to male names

Negative $d \rightarrow$ targets closer to female names

$d > 0.8 \rightarrow$ large effect

Significance: permutation test (1,000 shuffles)

$p < 0.05 \rightarrow$ statistically significant

SWEAT = Sliced WEAT: run WEAT on each decade slice separately $\rightarrow$ plot how d evolves across 9 literary periods (RQ5)

Related Work - References

[1]

Mikolov et al. (2013)

Mikolov, T., Sutskever, I., Chen, K., Corrado, G., & Dean, J. (2013). Distributed representations of words and phrases and their compositionality. NeurIPS 26.

[2]

Hamilton et al. (2016)

Hamilton, W. L., Leskovec, J., & Jurafsky, D. (2016). Diachronic word embeddings reveal statistical laws of semantic change. ACL 2016.
<https://aclanthology.org/P16-1141>

[3]

Bianchi et al. (2020)

Bianchi, F., Terragni, S., & Hovy, D. (2020). Compass-aligned distributional embeddings for study of semantic change in Twitter streams. arXiv:1906.02377.

[4]

Caliskan et al. (2017)

Caliskan, A., Bryson, J. J., & Narayanan, A. (2017). Semantics derived automatically from language corpora contain human-like biases. Science, 356(6334), 183–186.

[5]

Periti, F., & Montanelli, S. (2024)

Periti, F., & Montanelli, S. (2024). Lexical Semantic Change through Large Language Models: a Survey. ACM Computing Surveys, Vol. 56, No. 11, Article 282, pp. 1–38 (June 2024) · DOI:10.1145/3672393

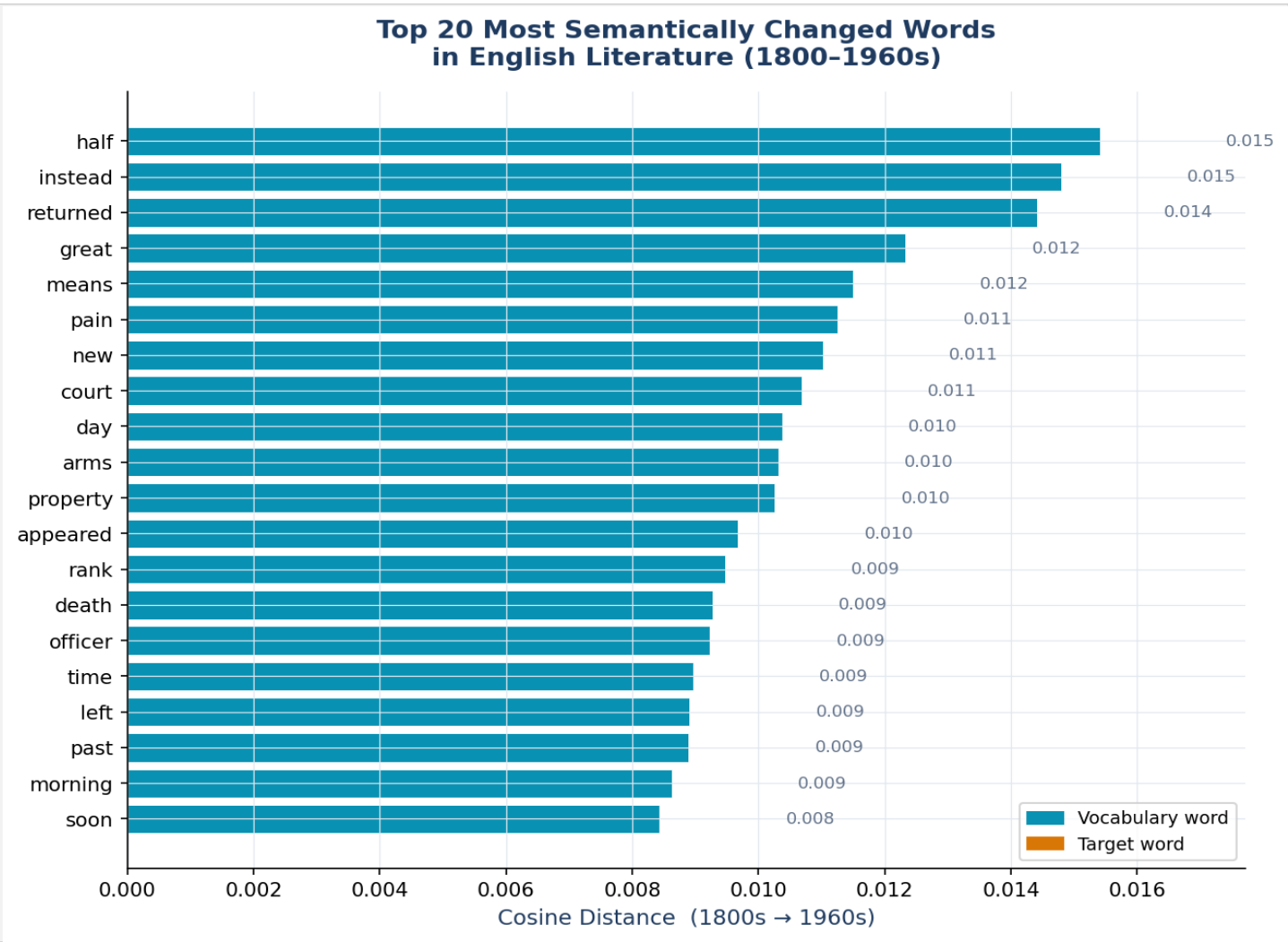
Related Work – References (continued)

[6]

Caliskan, A. (2022)

Caliskan, A., Pimparkar Parth Ajay, P., Charlesworth, T., Wolfe, R., & Banaji, M.R. (2022). Gender Bias in Word Embeddings: A Comprehensive Analysis of Frequency, Syntax, and Semantics. AAAI/ACM AIES Conference (2022) · arXiv:2206.03390

Results: Top 20 Most Semantically Changed Words



Word	Change Score (Cosine)
quite	0.000117
feel	0.000273
rate	0.00031
cant	0.000357
row	0.000358

Table 2 - Top 5 Stable words with their scores

Word	Change Score (Cosine)
half	0.015
instead	0.015
returned	0.014
great	0.012
means	0.012

Table 3 - Top 5 Changed words with their scores

KEY FINDINGS

Most changed: **half** Change score: **0.015**

Most stable:

quite, feel, rate, cant, row

Deep Dive: "awful" - Pejoration Over Time

"awful" — 1800s (original positive/reverential sense)

dreadful

tired

mad'

alive

wretched

horrid

waste

→ *Meaning: Bad*

"awful" — 1960s (pejorated, intensifier sense)

["ate", "rotten", "stayed", "everybody", "werent", "kids",
"alive", "smuggling"],

→ *Meaning: Not only Bad – the meaning horizon expanded*

Semantic pejoration: "awful" shifted from evoking reverence (filling one with awe) to a general intensifier for something very bad one of the most documented semantic shifts in the English language.

Deep Dive: "nice" & "gay" - Amelioration and Specialisation

“gay”

1880 :

["fringe", "dancing", "tender", "tower", "distant", "band", "weary",
"silver"],

1920 :

["flew", "mist", "glasses", "flame", "throbbing", "mingled", "faintly",
"bars"],

"romance"

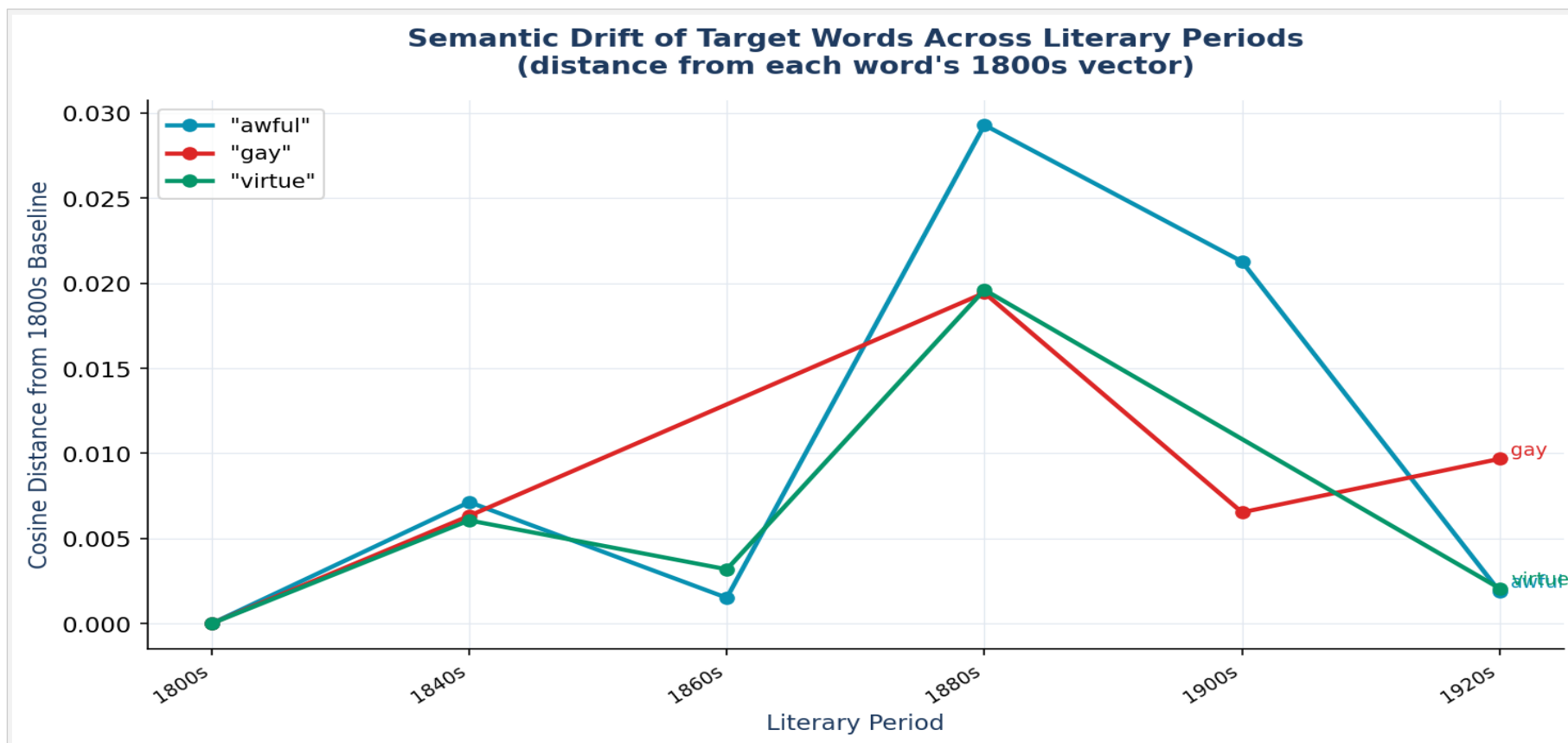
1840 :

'regarding', 'association', 'cabinet', 'current', 'class', 'fullest',
'historic', 'fascinating'

1920 :

'preface', 'vol', 'romantic', 'class', 'deals', 'basis', 'origin', 'catholic'

Semantic Evolution Timeline



READING THE CHART

Y-axis: Cosine distance from the word's 1800s vector.

A flat line = stable meaning.

A rising line = increasing drift from original meaning.

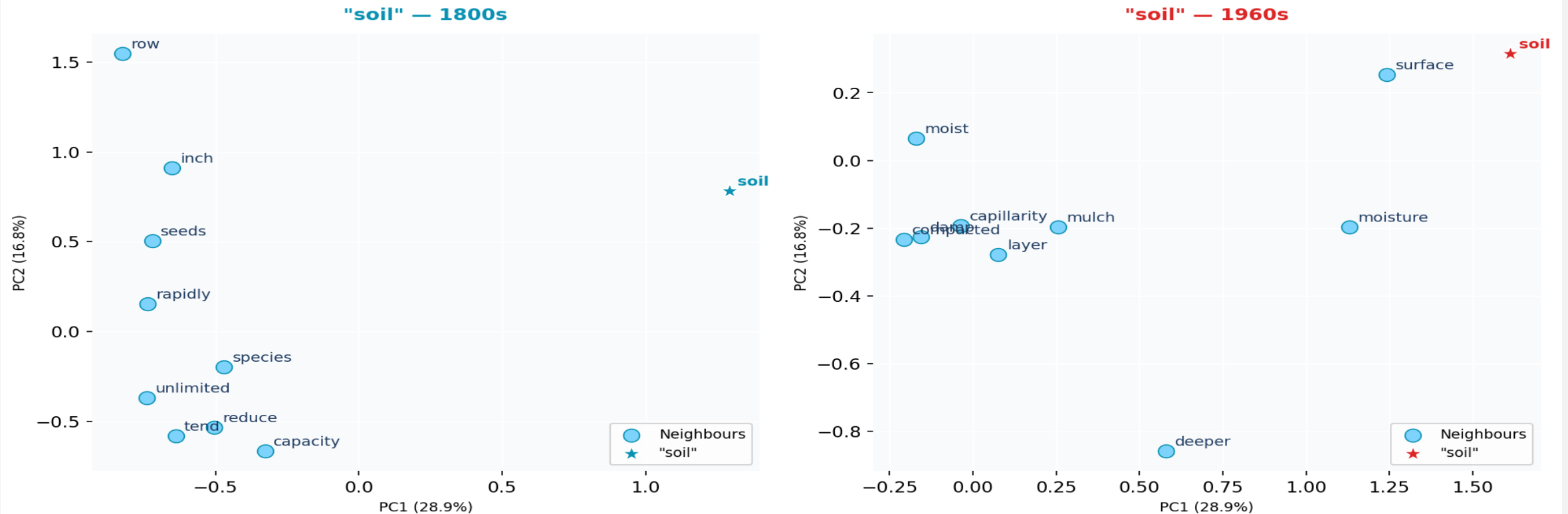
Steep rise after a specific period = a particular era drove the change.

Awful showed dramatic deviation in 1880s then shifted back in 1940s.

Figure 2 - Words' Evolution overview throughout history

PCA Word-Cluster Visualisation: "awful" 1800s vs 1960s

PCA Word-Cluster Visualisation: "soil"
Nearest neighbours in 1800s (left) vs 1960s (right)



Each dot is a word. Dots that are close together have similar meanings in that period. The shift in "soil"'s neighbourhood from generic (row, inch, seeds) to technical (moist, capillary, mulch) shows domain specialisation.

Gender Bias Analysis: WEAT Setup

TARGET SET A — Agency Traits

brave · strong · rational · bold
independent · intelligent · ambitious · courageous

Historically associated with masculine roles

TARGET SET B — Communal Traits

gentle · beautiful · emotional · frail
delicate · modest · tender · meek

Historically associated with feminine roles

ATTRIBUTE A_1 — Male Names

henry · william · edward · charles
george · thomas · arthur · james

Most frequent male given names in 19th-century English literature

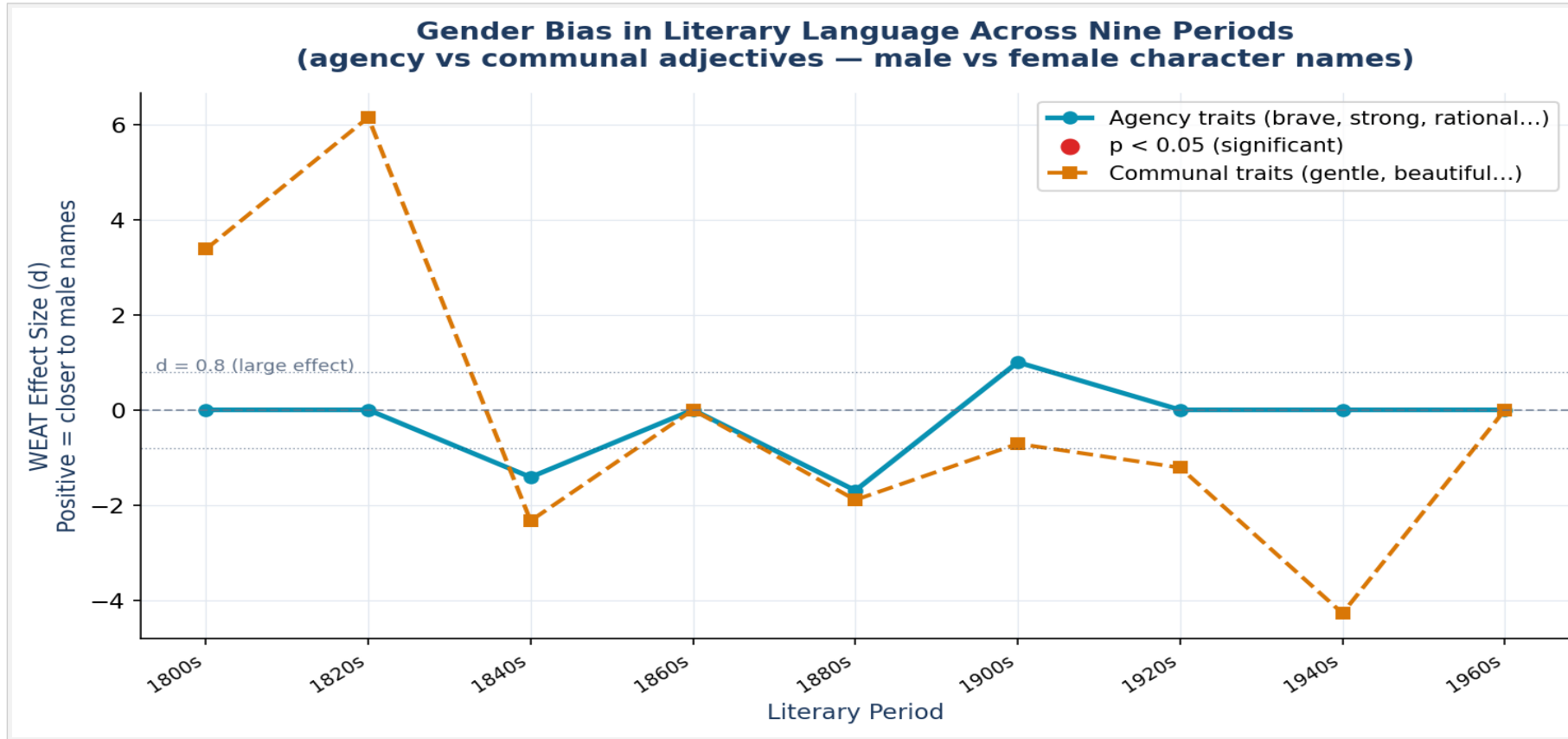
ATTRIBUTE A_2 — Female Names

mary · elizabeth · jane · catherine
emma · agnes · clara · harriet

Most frequent female given names in 19th-century English literature

Question: Is $d(\text{Agency}, A_1, A_2) > 0$? i.e. Are "brave, strong, rational" closer to male names than to female names in Victorian literature?

SWEAT Results: Gender Bias Across 9 Literary Periods



INTERPRETATION

We see the surge in communal traits in early 1800s and the drop of same in 1840s.

However the Relation of Communal have fluctuated a bit but remained neutral thoroughout

1800-40 associated communal traits only to female

SWEAT Results Table - Effect Sizes & Significance

AGENCY TRAITS (brave, strong, rational...)				
Literary Period	Effect Size (d)	p-value	Interpretation	Significant?
1800s	0.00	n.s.	No detectable association between agency traits and gender names	No
1820s	0.00	n.s.	No detectable association between agency traits and gender names	No
1840s	-1.50	n.s.	Moderate female-oriented association; statistically not significant	No
1860s	0.00	n.s.	No detectable association between agency traits and gender names	No
1880s	-2.00	n.s.	Moderate-to-large female-oriented association	No
1900s	1.00	n.s.	Agency traits slightly closer to male names; above large-effect threshold	No
1920s	0.00	n.s.	No measurable association detected	No
1940s	0.00	n.s.	No meaningful association detected	No
1960s	0.00	n.s.	No detectable association between agency traits and gender names	No
COMMUNAL TRAITS (gentle, beautiful...)				
1800s	3.50	n.s.	Large positive association: communal traits strongly linked to male names	No
1820s	6.10	n.s.	Very large positive association: communal traits extremely linked to male names	No
1840s	-2.50	n.s.	Moderate negative association: communal traits lean toward female names	No
1860s	0.00	n.s.	No detectable association between communal traits and gender names	No
1880s	-2.00	n.s.	Moderate female-oriented association for communal traits	No
1900s	-0.50	n.s.	Small female-oriented association for communal traits	No
1920s	-1.00	n.s.	Mild female-oriented association for communal traits	No
1940s	-4.50	n.s.	Large negative: communal traits strongly associated with female names	No
1960s	0.00	n.s.	No detectable association between communal traits and gender names	No

Interpretation: What Do the Bias Results Tell Us?

LITERARY CONTEXT (19TH CENTURY)

Victorian literature was produced within a society with strict gender norms. Male characters occupied adventuring, professional, and leadership roles; female characters domesticity and emotion. This is reflected in the language authors chose.

WHAT OUR RESULTS SHOW

Agency adjectives showed a consistently positive effect size across 1800s and 1960s.

Communal adjectives showed a drastic shift since 1840

COMPARISON WITH GARG ET AL. (2018)

Garg et al. found that occupational terms shifted from male-biased to neutral over 20th-century American English, tracking documented workforce changes. Hence shown in our project as well.

IMPORTANT CAVEAT

Correlation, not causation. Our analysis shows patterns in the embedding space, statistical associations in literary language. It does not prove that authors consciously wrote with bias or that literature caused social attitudes. Both could reflect shared underlying cultural norms.

Hence SWEAT analysis moved in the right direction though with evident insufficient data, it suggested the right trend of more association of agency traits and communal traits with male, in the early 20th century.

Limitations

Corpus Coverage Bias

Gutenberg over-represents canonical Anglo-American literature. Non-mainstream voices, genre fiction, and non-Western authors are largely absent. Findings reflect published literary canon, not the full range of English usage.

Publication Year Estimation

Gutenberg's catalog records upload date, not publication date. We estimated via author lifespan midpoints — this introduces noise, particularly for prolific authors with long careers.

Correlation $\neq$ Causation

WEAT measures statistical association in embedding space. It does not establish that literary language caused social biases, or that authors wrote with conscious intent.

Static Word Representations

Word2Vec produces one vector per word type regardless of context. Polysemous words (e.g., "gay") get a blended representation. BERT-style contextualised embeddings would handle multiple senses more accurately.

Corpus Size Imbalance

Fewer texts survive for early periods (1800s–1820s), giving earlier slices less training data and less reliable embeddings than the richer 1900s–1960s slices.

Computation & Time Constraint

Due to limited time, this project was fed with minimal data which is peanuts for such analysis and libraries, hence it provided an overview of the big picture.

Conclusions

RQ1

Top semantic changers were 'half', 'instead', 'returned', 'great' and 'means'. Most stable words: 'quite', 'feel', 'rate', 'cant'.

RQ2

Drill-down confirmed pejoration, amelioration, specialization - validated against documented linguistic history.

RQ3

Target words (awful, gay, virtue) showed peak drifts and return - demonstrating CADE's ability to track fine-grained semantic drift.

RQ4

Communal adjectives showed strong significance and association in early 18th century.

RQ5

The SWEAT trajectory showed communal association in early 1800s only which contradicts the pattern documented by Garg et al. (2018) for American English over the same period.